



Department of Information Science & Engineering QUESTION BANK

Semester: V

Section: 'A' & 'B'

Subject Name & Code: Theory of Computation [BCS503]

MODULE-1

- 1) a. Define a DFA and language accepted by it.
b. Define (i) Alphabet (ii) Power of an alphabet (iii) String (iv) Language v) Length of string
iv) Concatenation of strings v) string reversal
- 2) Write the DFSM for the following languages and List a string belonging to the L and a string not belonging to L.
 - i) The set of all strings ending with abb and show the sequence of moves made by the DFA for the strings ababb and aab.
 - ii) The set of all strings not containing substring abb.
 - iii) The set of all strings with number of a's is multiple of 3 and number of b's is multiples of 5.
 - iv) Set of all strings such that number of 1's is even and number of 0's is a multiple of 3 on $\Sigma = \{0,1\}$.
 - v) Design a DFSM for the $L = \{w : |w| \bmod 5 \geq |w| \bmod 3\}$ on $\Sigma = \{0,1\}$.
 - vi) $L = \{w \in \{0,1\}^* : w \text{ has } 001 \text{ as a substring}\}$
 - vii) To accept string of 0's, 1's and 2's beginning with a '0' followed by odd number of 1's and ending with a '2' over $\Sigma = \{0,1,2\}$.
 - viii) To accept the string of a's and b's ending with ab or ba.
 - ix) Set of all strings with exactly three consecutive a's
 - x) Obtain a DFA to accept the language $L = \{w : |w| \bmod 5 \neq 0\}$ on $\Sigma = \{a\}$
- 3) Design a DFA which accepts set of all strings of 0's and 1's begins with 1 that when interpreted as binary integer is multiple of 5. ex: 101, 1010, 1111 are in the language and 0, 0101 are not.
- 4) Construct a DFA to accept decimal strings divisible by 3
- 5) Build a Nondeterministic FA for each of the following languages. List a string belonging to the L and a string not belonging to L.
 - i. $L = \{w | w \in ababn \text{ or } aban \text{ where } n \geq 0\}$
 - ii. To accept the string of a's and b's ending with ab or ba.
- 6) Obtain a DFA to accept strings of a's and b's having
 - A. Even number of a's and Even number of b's
 - B. Even number of a's and Odd number of b's
 - C. Odd number of a's and Even number of b's
 - D. Odd number of a's and Odd number of b's
- 7) Define ϵ -NFA and ϵ – CLOSURE. Obtain an NFA for the following regular expressions.
 - i) $a^* + b^* + c^*$ ii) $(a+b)^* aab (a+b)^*$ iii) $(0+1)^* 1(0+1)^*$
- 8) Obtain the DFA for the following NFA using subset Evaluation method

δ	0	1
$\rightarrow p$	$\{p, q\}$	$\{p\}$
q	$\emptyset$	$\{r\}$
*r	$\{p, r\}$	$\{q\}$

- 9) Design NFA's to recognize the following set of strings:
 - i) lab, cab and dab. Assume the alphabet is $\{a, b, c, d, l\}$
 - ii) 1101, 101 and 111. Assume the alphabet is $\{0, 1\}$

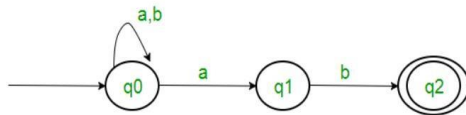
- Convert each of your NFA's from i) and ii) to DFA's.
10) Convert the following ϵ -NFA to DFA



- 11) Consider the following ϵ -NFA

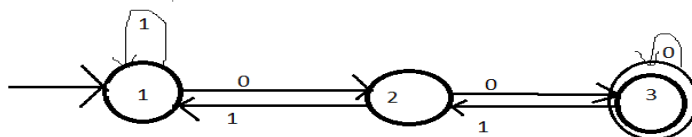
δ	ϵ	0	1
$\rightarrow p$	{r}	{q}	{p,r}
q	$\emptyset$	{p}	$\emptyset$
*r	{p,q}	{r}	{p}

- Compute -Closure for each state.
 - Convert automaton to DFA.
 - Give the set of all strings of length 3 or less accepted by the automaton.
- 12) Convert the following NFA to its equivalent DFA



MODULE-2

- Define Regular expression and write the regular expression for the following:
 - $L = \{a^{2n}b^{2m+1} \mid m \geq 0, n \geq 0\}$
 - $L = \{a^n b^m \mid m+n \text{ is even}\}$
 - $L = \{a^n b^m \mid m \leq 3, n \geq 4\}$
 - $L = \{a^n b^m \mid m \geq 1, n \geq 1, nm \geq 3\}$
 - String of a's and b's having even length of a's followed by odd length of b's.
 - String of a's and b's whose length is either even or multiples of 3 or both.
- Obtain a regular expression to accept the words with two (or) more letters but beginning and ending with the same letter over the input {a,b}.
- Obtain the regular expression to accept strings of a's, b's, c's such that 4th symbol from the right is a and ends with b.
- Give the regular expression for the following languages
 - $L = \{w \mid |w| \bmod 4 = 0\}$ on $\Sigma = \{a,b\}$.
 - $L = \{w \mid w \text{ contains even number of 0's followed by odd number of 1's}\}$.
 - $L = \{w \in \{a,b\}^* \mid w \text{ has exactly one pair of consecutive a's}\}$.
 - Set of all strings not ending in substring 'ab' over $\Sigma = \{a,b\}$.
- Write the regular expression for the following
 - $\{w \in \{a,b\}^* \mid \text{every a in w is immediately preceded and followed by b}\}$.
 - $\{w \in \{a,b\}^* \mid w \text{ does not end in ba}\}$.
 - $\{anbm \mid m \geq 1, n \geq 1, nm \geq 3\}$ (Feb 2021)
 - $\{w \in \{0,1\}^* \mid w \text{ has 001 as a substring}\}$.
 - $\{w \in \{a,b\}^* \mid w \text{ has both aa and bb as substrings}\}$.
 - $\{w \in \{0,1\}^* \mid \text{none of the prefixes of w ends in 0}\}$.
 - $\{w \in \{a,b\}^* \mid \#a(w) = 3\}$.
 - $\{w \in \{a,b\}^* \mid \#a(w) \leq 3\}$.
 - $\{w \in \{a,b\}^* \mid w \text{ contains no more than two occurrences of the substring aa}\}$.
- Prove that, for every regular expression there exists a finite automaton which accepts the same language accepted by regular expression.
- Convert the following automaton to regular expression using state elimination method





- 8) State and prove pumping lemma theorem for regular languages.
- 9) Apply pumping lemma for the following languages and prove that they are not Regular.
 - i) $L_1 = \{ a^n b^n \mid n \geq 0 \}$
 - ii) $L_2 = \{ w \mid n_a(w) < n_b(w) \}$
 - iii) $L_3 = \{ a^n \mid n > 0 \}$
 - iv) $L_4 = \{ a^n \mid n = k^2, n \geq 1 \}$
- 10) Prove that if $L = L(A)$ for some DFA A, then there is a regular expression R such that $L = L(R)$
- 11) Define distinguishable and indistinguishable states. Minimize the following DFAs using table filling algorithm:
 - i)
 - ii)

δ	0	1
$\rightarrow A$	B	E
B	C	F
*C	D	H
D	E	H
E	F	I
*F	G	B
G	H	B
H	I	C
*I	A	E

σ	0	1
$\rightarrow A$	B	A
B	A	C
C	D	B
*D	D	A
E	D	F
F	G	E
G	F	G
H	G	D

iii)

σ	a	b
$\rightarrow q_0$	q_1	q_2
q_1	q_1	q_3
q_2	q_1	q_2
q_3	q_1	q_4
* q_4	q_1	q_2

- 12) Prove that regular languages are closed under union, intersection, concatenation, complementation, reverse and homomorphism.
- 13) If L_1 and L_2 are regular then prove that $L_1 \cup L_2$ is also regular.
- 14) Consider the homomorphism from the alphabet $\{0,1,2\}$ to $\{a,b\}$ defined by $h(0)=ab$, $h(1)=b$, $h(2)=aa$.
 - (i) What is $h(2201)$?
 - (ii) If L is language 1^*02^* , what is $h(L)$?
 - (iii) If L is language $(ab+abab)^*bab$, what is $h^{-1}(L)$?

MODULE-3

- 1) Define Context Free Grammar. Design CFG for the following Languages.
 - a) $L = \{ a^n b^{n+3} \mid n \geq 3 \}$
 - b) $L = (01)^n \mid n \geq 1$
 - c) Let $\Sigma = \{a, b\}$ obtain a grammar G generating set of all palindromes over Σ .
 - d) Let $\Sigma = \{a, b\}$ obtain a grammar G generating set of all non-palindromes over Σ .
 - e) $L = \{ a^n b^n \mid n \geq 1 \}$
 - f) $L = \{ a^n b^m \mid n \geq 0, m > n \}$ **(June/Jul 2023, 18 scheme)**
 - g) $L = \{ a^n b^{2n} \mid n \geq 0 \}$ **(Jan/Feb 2021, 18 scheme)**
 - h) $L = \{ ww^R \mid w \in \{a,b\}^* \}$
 - i) $L = \{ 0^m 1^m 2^n \mid m \geq 1, n \geq 0 \}$
 - j) $L = \{ a^m b^n c^k \mid m+2n=k \text{ \& } m, n \geq 0 \}$
 - k) $L = \{ a^n b^{n-3} \mid n \geq 3 \}$ **(Dec 2018/Jan 2019, 15 scheme)**

- l) $L = \{w : |w| \bmod 3 \text{ not equal to } |w| \bmod 2\}$ on $w \in \{a,b\}^*$
- m) $L = \{\text{strings of a's and b's with equal no. of a's and b's}\}$ Also show the moves made by the PDA for the string "aabbab". **(Aug/Sep 2020,17 Scheme) (Aug/Sep 2020,17 Scheme)**
- n) $L_2 = \{a^n b^m c^k : m = n + k\}$ over $\Sigma = \{a, b, c\}$ **(Feb/March 2022,18 scheme)**
- o) $L_1 = \{w : |w| \bmod 3 = 0\}$ over $\Sigma = \{a\}$
- p) $L = \{0^m 1^m 2^n \mid m \geq 0, n \geq 0\}$
- q) $L = \{0^m 1^m 2^n \mid m \geq 0, n \geq 0\}$
- r) $L = \{a^i b^j \mid i=j+1\}$ **(Aug/Sep 2020,17 Scheme)**
- s) $L = \{W \mid n_a(W) = n_b(W) \mid n \geq 1\}$
- t) $L = \{a^{(2n+1)} b^m c^m c^{2n} \mid n, m \geq 0\}$ **(July/Aug 2022,15 scheme)**
- u) $L = \{a^i b^j c^k : i+j=k, i \geq 0, j \geq 0\}$
- v) $L = \{a^n b^m c^k : n+2m=k\}$ **(Jan/Feb 2021,15 scheme) (Jan/Feb 2023,18 scheme)**
- w) $L = \{a^i b^j \mid i \neq j, i \geq 0, j \geq 0\}$ **(June/Jul 2023,18 scheme)**
- x) $L = \{a^n b^{n+1} \mid n \geq 0\}$ **(June/Jul 2023,17 scheme)**
- 2) What is NPDA? Design NPDA for Language $L = \{a^n b^n \mid n \geq 1\}$. Draw transition diagram. Write sequence of moves made by NPDA to accept the string aaabbb. **(2022, 2018,19,23,17)**
- 3) Define PDA. Obtain PDA to accept the language $L(M) = \{wcw^R \mid w \in (a+b)^*\}$, where w^R is the reverse of w by a final state. And show the moves made by PDA for the string "aabcbbaa" and "abacbbba". **(Jan/Feb 2023,18 scheme) (June/Jul 2023,18 scheme) (Jul/Aug 2022)**
- 4) Construct a PDA to accept the language $L = \{ww^R \mid w \in \{a, b\}^*\}$. Draw the graphical representation of this PDA. Show the moves made by this PDA for the string aabbbaa. **(Dec 2017, Jan 2018) (June/Jul 2023,17 scheme) (Jan/Feb 2023,18 scheme)**
- 5) Design a PDA for the language that accepts the string with $n_a(w) < n_b(w)$, where $w \in \{a+b\}^*$ and show the instantaneous description of the PDA on input abbab. **(Jan/Feb 2021,18 scheme)**
- 6) Design a PDA for the language that accepts the string with $n_a(w) = n_b(w)$, where $w \in \{a+b\}^*$ and show the instantaneous description of the PDA on input abbaba.
- 7) Build a PDA to accept delimiters or balanced parentheses having parentheses $\{ , (,) , \}$
- 8) Obtain a CFG for PDA M with the transitions:
 $\delta(q_0, a, Z) = (q_0, AZ)$
 $\delta(q_0, b, A) = (q_0, AA)$
 $\delta(q_0, a, A) = (q_1, \epsilon)$ **(Jan/Feb 2021,15 scheme)**
- 9) Convert the following CFG to PDA:
 $S \rightarrow aABB \mid aAA$
 $A \rightarrow aBB \mid a$
 $B \rightarrow bBB \mid A$
 $C \rightarrow a$ **(Dec 2017, Jan 2018) (Jul/Aug 2022,17 scheme)**
- 10) Convert the following grammar into equivalent PDA:
 $S \rightarrow aABC$
 $A \rightarrow aB \mid a$
 $B \rightarrow bA \mid b$
 $C \rightarrow a$ **(Dec 2018/Jan 2019,15 scheme) (June/Jul 2023,17 scheme)**
- 11) Consider the grammar G with production Obtain the left most derivation, right most derivation and parse tree for some string for the following CFG.
- a) $S \rightarrow AbB$
 $A \rightarrow Aa \mid \epsilon$
 $B \rightarrow aB \mid bB \mid \epsilon$
 For the string aaabab.
- b) $S \rightarrow aB \mid bA$
 $A \rightarrow aS \mid bAA \mid a$
 $B \rightarrow bS \mid aBB \mid b$
 For the string aaabbabba.
- c) $E \rightarrow E + E$
 $E \rightarrow E - E$

$E \rightarrow E * E$
 $E \rightarrow E / E$
 $E \rightarrow E \wedge E$
 $E \rightarrow id$

For the string $id+id^{\wedge}id$

12) Prove that the given grammars below are ambiguous.

- a) $S \rightarrow aS \mid X$
 $X \rightarrow aX \mid a$
- b) $S \rightarrow aB \mid bA$
 $A \rightarrow aS \mid bAA \mid a$
 $B \rightarrow bS \mid aBB \mid b$
- c) $S \rightarrow iCtS \mid iCtSeS \mid a$
 $C \rightarrow b$

MODULE 4

1) Eliminate ϵ , unit and useless productions from the following grammar:

$S \rightarrow AaA \mid CA \mid BaB$
 $A \rightarrow aaBa \mid CDA \mid aa \mid DC$
 $B \rightarrow bB \mid bAB \mid bb \mid aS$
 $C \rightarrow Ca \mid bC \mid D$
 $D \rightarrow bD \mid \epsilon$

2) What is a unit production? Begin with the grammars

- (i) $S \rightarrow ABC \mid BaB$
 $A \rightarrow aA \mid BaC \mid aaa$
 $B \rightarrow bBb \mid a \mid D$
 $C \rightarrow CA \mid AC$
 $D \rightarrow \epsilon$
- (ii) $S \rightarrow aAa \mid bBb \mid \epsilon$
 $A \rightarrow C \mid a$
 $B \rightarrow C \mid b$
 $C \rightarrow CDE \mid \epsilon$
 $D \rightarrow A \mid B \mid ab$

Eliminate ϵ -productions.

Eliminate any unit production in the resulting grammar.

Eliminate any useless symbols in the resulting grammar.

3) Remove useless symbols from the following grammar:

$S \rightarrow aA \mid B$
 $A \rightarrow aA \mid a$
 $B \rightarrow bB$
 $D \rightarrow ab \mid Ea$
 $E \rightarrow ac \mid d$

4) What is Chomsky Normal Form of CFG? Convert the following grammar to CNF. **(2022,18 scheme)**

$S \rightarrow ABC \mid BaB$
 $A \rightarrow aA \mid BaC \mid aaa$
 $B \rightarrow bBb \mid a \mid D$
 $C \rightarrow CA \mid AC$
 $D \rightarrow \epsilon$

Eliminate ϵ -productions, Unit productions and useless symbols if any before conversion.

5) Obtain the following grammar in CNF

$S \rightarrow ABC$
 $A \rightarrow aC \mid D$
 $B \rightarrow bB \mid E \mid A$
 $C \rightarrow Ac \mid E \mid Cc$
 $D \rightarrow aa$ **(Aug/Sep 2020,17 Scheme)**

6) Define CNF. Design PDA for the language $L = \{0^n b^m 1^{2n} \mid n, m \geq 0\}$. Write Instantaneous Description for the string 0bb11. **(July/Aug 2022,15 scheme)**

7) Convert the given grammar into CNF :

$S \rightarrow 0A \mid 1B$
 $A \rightarrow 0AA \mid 1S \mid 1$
 $B \rightarrow 1BB \mid 0S \mid 0$ **(June/Jul 2023,18 scheme) (Jul/Aug 2022,17 scheme)**

8) Simplify the grammar by removing productive and unreachable symbols:

$S \rightarrow AB \mid AC$
 $A \rightarrow aA b \mid \epsilon$
 $B \rightarrow bB$
 $C \rightarrow bCa$
 $D \rightarrow AB$ **(Jul/Aug 2022,18 scheme)**

- 9) Convert the following grammar to CNF:
 $S \rightarrow ASB$
 $A \rightarrow aAS \mid a$
 $B \rightarrow SbS \mid A \mid bb$ **(Jul/Aug 2022,18 scheme) (Jan/Feb 2023,18 scheme)**
- 10) What is CNF and GNF? Convert the following grammar into GNF:
 $S \rightarrow AA \mid a$
 $A \rightarrow SS \mid b$ **(Jan/Feb 2021,18 scheme)**
- 11) Convert the grammar into Chomsky Normal Form (CNF)
 $S \rightarrow aAD,$
 $A \rightarrow aB \mid bAB,$
 $B \rightarrow b,$
 $D \rightarrow d$ **(June/Jul 2023,17 scheme)**
 ii) $S \rightarrow OA \mid 1B$
 $A \rightarrow OAA \mid 1S \mid 1$ **(June/Jul 2023,17 scheme)**
 iii) List out the simplification techniques of CFG. Explain PDA and obtain an equivalent PDA for CFG given below: $S \rightarrow a \mid b \mid a s b \mid b s a$ **(July/Aug 2022,15 scheme)**
- 12) Design a FSM for the following regular grammar and obtain corresponding regular expression.
 a) $S \rightarrow bS \mid aT \mid \epsilon$ b) $S \rightarrow aT$
 $T \rightarrow bS$ $T \rightarrow bT \mid a \mid aW$
 $W \rightarrow \epsilon \mid aT$
- 13) Eliminate left recursion from the following grammars.
 a) $E \rightarrow E+T \mid T$ b) $A \rightarrow Aab \mid AbB \mid ab \mid aa$ c) $S \rightarrow Ab \mid a$
 $T \rightarrow T*F \mid F$ $B \rightarrow bB \mid b$ $A \rightarrow ASa$
 $F \rightarrow (E) \mid id$
- 14) Convert the following grammar into an equivalent unambiguous grammar
 a) $S \rightarrow ABA$ b) $E \rightarrow E+E \mid E-E$
 $A \rightarrow Aa \mid \epsilon$ $E \rightarrow E*E \mid E/E$
 $B \rightarrow bB \mid \epsilon$ $E \rightarrow (E) \mid I$
 $I \rightarrow a \mid b \mid c$

MODULE-5

- 1) Design TM for WCW^R over $\Sigma = \{0, 1\}$. Write transition diagram, and ID for $w = 101C101$ **(Feb/March 2022,18 scheme)**
- 2) Design Turning machine to accept the Language $L = \{0^n 1^n 2^n \mid n \geq 0\}$. Draw the transition diagram. Write sequence of moves made by TM for string 001122. **(Feb/March 2022,18 scheme)**
- 3) Design and construct TM to accept the language $L = \{W \mid W \text{ is balanced parenthesis}\}$. Show the moves made by TM for string $W = (() ())$. **(July/Aug 2022,15 scheme)**
- 4) Design a Turing Machine for the given languages
 i) $L = \{0^n 1^n \mid n \geq 0\}$ **(Jul/Aug 2022,17 scheme)**
 ii) $L = \{a^n b^n c^n \mid n \geq 1\}$. Draw the transition diagram. Show the moves made by TM for the string aabbcc. **(Dec 2017, Jan 2018) (June/Jul 2023,18 scheme) (Jul/Aug 2022,17 scheme)**
 iii) $L = \{0^n 1^n \mid n \geq 1\}$. Draw the transition diagram. Show the moves made by the machine for the string 000111, 1001, 0011 and 00111. **(Jul/Aug 2022,17 scheme) (Jan/Feb 2023,18 scheme)**
 iv) $L = \{1^n 2^n 3^n \mid n \geq 1\}$. Show that the string 112233 is accepted by ID. **(Jul/Aug 2022,18 scheme)**
- 5) Explain the working principle of Turning Machine with diagram. Design a TM to accept strings formed on $\{0, 1\}$ and ending with 000. Write transition diagram and ID for $w = 101000$. **(2021,18)**
- 6) Define TM and design a TM for $L = \{w \mid w \in (0, 1)^* \text{ containing the substring } 001\}$. Write transition diagram and show the moves made by the TM for input string 10010. **(Jan/Feb 2023,18 scheme)**
- 7) Design a TM to accept strings of a's and b's ending with ab or ba. **(2018,2019,15)**
- 8) Explain halting problem in Turing machine. Prove that $HALT_{TM}(M, W) = \{ \langle M, w \rangle \mid \text{The Turing machine } M \text{ halts on input } w \}$ is undecidable. **(June/Jul 2023,18)**
- 9) Write Short notes on
 i) Post correspondence problem with respect to TM ii) Quantum Computers iii) Church Turing Thesis
 iv) Linear bound Automation v) variants of Turing machine vi) Halting problem vii) decidability viii) undecidable language. ix)) Classes of P and NP

